Toy Linear Algebra

1. Vector spaces / 1.1. The zero subspace

This is an original test fixture, not an extract from a textbook.

Let V be a vector space over a field F, and let U = {0}.

The zero vector belongs to U.

The sum of two elements of U is zero and belongs to U.

For every scalar a in F, a times the zero vector is zero.

Thus U is closed under addition and scalar multiplication.

Together these conditions show that U is a subspace of V.